

The Anatomy of the Treasury Market

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Background

- ▶ Following Koijen and Yogo (KY, 2019) and Gabaix and Koijen (2023), a growing literature uses holdings/flows to estimate asset demand systems
- ▶ Allows *decomposing* returns into contribution by different investors

$$\Delta p = \Delta p^{(\text{HF})} + \Delta p^{(\text{PF})} + \Delta p^{(\text{IA})} + \Delta p^{(\text{Dealers})} + \Delta p^{(\text{Foreign})} \dots$$

- ▶ Holdings/flows data provides incremental information for prices if aggregate demand is not perfectly elastic.
- ▶ Alternative view: Demand is very elastic and only common factors drive variation in prices (→ representative agent)

This Paper: Summary

- ▶ Propose and estimate a demand system for the **Treasury Market**.
- ▶ Setting: N investors, T observations of trades
→ **Goal**: Extract *who* moves yields out of this panel.
- ▶ Two equation model
 1. Investor i 's demand has elasticity ζ_i and shift $d_{i,t}$.

$$\Delta q_{i,t} = \zeta_i \Delta p_t + d_{i,t}$$

2. Market clearing, $\sum_i S_i \Delta q_{i,t} = 0$, implies

$$\Delta p_t = \frac{1}{\zeta_S} \sum_i S_i d_{i,t}$$

- ▶ Equipped with the estimated demand system, the paper **dissects** the treasure market

$$\Delta p_t = \Delta p_t^{(\text{Fed})} + \Delta p_t^{(\text{R.o.W.})} + \Delta p_t^{(\text{Fin. Inst.})} + \Delta p_t^{(\text{HH})} + \dots$$

into investor-specific determinants given by their (scaled) demand shifts $\Delta p_t^{(i)} = d_{i,t} \frac{S_i}{\zeta_S}$.

Comments

- ▶ **Great!** If we want to understand the treasury market, probably useful to look at who is buying and selling treasuries.
- ▶ New identification method, rich empirical findings. Big step forward for understanding movements in US treasury yields systematically.
- ▶ One of the key innovations: Optimal GIV estimator with heterogeneous (investor-specific) elasticities.
- ▶ Comments
 1. Identifying heterogeneous elasticities and sensitivity to misspecification.
 2. Limitations of framework (strict separation of slope/shift).
 3. How much incremental information is there in treasury flows?

Identifying heterogenous elasticities

<i>Sector</i>	<i>S(%)</i>	ζ	ζ Share (%)
Aggregate		1.03 (0.77, 1.3)	100.0
Households	5.74	10.54 (5.33, 15.76)	58.55
RoW	44.45	0.42 (0.23, 0.61)	17.8
Fed	22.08	0.42 (0.11, 0.74)	9.28
Mutual Funds	6.75	0.6 (0.13, 1.08)	3.95
Insurance	2.59	1.08 (0.76, 1.4)	2.7
Banks	5.26	0.52 (0.28, 0.76)	2.61
Dealers	0.81	2.22 (-6.45, 10.88)	1.73

- ▶ Great heterogeneity in elasticities!
- ▶ Identifying investor-specific elasticities ζ_i and demand shocks $d_{i,t}$ are two sides of the same coin.

Why estimating **investor-specific** curves is non-trivial

- ▶ Simplest case: Run $i = 1, \dots, N$ different OLS regressions (1 regression for each investor).
 - ▶ Closed-form OLS estimate: $\hat{\zeta}_i = \frac{\text{cov}(\Delta q_i, \Delta p)}{\text{var}(\Delta p)}$
 - ▶ Size-weighted sum $\hat{\zeta}_S = \sum_i S_i \frac{\text{cov}(\Delta q_i, \Delta p)}{\text{var}(\Delta p)} = 0$. **No equilibrium!**
- ▶ Estimation with single instrument z for everyone:
 - ▶ Closed-form IV estimate: $\hat{\zeta}_i = \frac{\text{cov}(\Delta q_i, z)}{\text{cov}(\Delta p, z)}$
 - ▶ Size-weighted sum $\hat{\zeta}_S = \sum_i S_i \frac{\text{cov}(\Delta q_i, z)}{\text{cov}(\Delta p, z)} = 0$. **Still no equilibrium!**
- ▶ Need **investor-specific** instruments z_i .
 - ▶ Common instruments z , or equivalently controlling for cool macro factors, PCs, etc. alone is not enough.
 - ▶ Behind the scenes, the proposed GIV estimator constructs investor-specific (leave-1-out) instruments $z_i = \sum_{j \neq i} S_j u_j$

Identifying heterogenous elasticities

- ▶ **Goal:** Construct instrument for each investor by summing the demand shocks of others
 $z_i = \sum_{j \neq i} S_j u_j$.
- ▶ Requires eliminating elasticity term from $\Delta q_{i,t} = \zeta_i \Delta p_t + u_{i,t}$.

1. Homogenous Elasticities ζ (Standard GIV):

Cross-sectional demeaning kills elasticity term: $\hat{u}_{i,t} = \Delta q_{i,t} - \Delta \bar{q}_t$

- ▶ Simple and intuitive ✓, but can't get investor-specific elasticities.
- ▶ How much do your decompositions change with this simplified estimator? Probably a lot given the heterogeneity in elasticities...

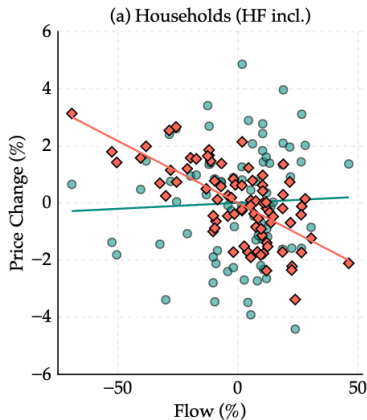
2. Heterogenous Elasticities ζ_i (GK Section D.9.1)

Guess ζ_i , construct z_i , and iterate $E[u_i z_i] = 0$ until convergence.

- ▶ Looks similar to your estimator. What's the main difference?
- ▶ Do you get the same results?

Sensitivity to Misspecification

- ▶ Households are the most elastic investor group $\hat{\zeta}_{HH} = 10$



- ▶ But their demand is positively correlated with prices! At first glance, they are liquidity *taking*.
- ▶ The instrument seems to be doing *a lot* of work.

Sensitivity to Misspecification: Event Studies

- ▶ Say I add one more hypothetical quarter of data:
2025 Q2 → Yields finally drop by 100bp.²
- ▶ This will have little effect on the estimated elasticities $\hat{\zeta}_i$
(which are estimated over 20 years).
- ▶ Inferred demand shocks for each investor:

$$\hat{u}_{i,t} = \Delta q_{i,t} + \hat{\zeta}_i \Delta p_t$$

- ▶ For simplicity assume that there is little trading in Flow of Funds data $\Delta q_{i,t} \approx 0$.
- ▶ In this case: return decomposition entirely driven by elasticity distribution $\hat{u}_{i,t} = \hat{\zeta}_i \Delta p_t$

$$1 \approx \underbrace{S_{HH} \hat{\zeta}_{HH}}_{56 \%} + \underbrace{S_{RoW} \hat{\zeta}_{RoW}}_{18 \%} + \underbrace{S_{Fed} \hat{\zeta}_{Fed}}_{9 \%} + \dots^3$$

- ▶ When there is little trading, elasticities do all the heavy lifting. Average (20 year) elasticities less ideal for event studies?

²All it took was a little trade war.

³Assuming that $\zeta_S \approx 1$

How much incremental information is there in treasury holdings/flows beyond macro variables?

- ▶ Straightforward answer: If markets are inelastic → a lot!
Difficult to test...
- ▶ Alternative suggestion: Does flow data improve link between macro variables and yield changes?
 - ▶ Simple OLS Regression:

$$\Delta p_t = \alpha + \beta_v \Delta \text{VIX}_t + \beta_F \Delta \text{FF-Rate}_t + \dots + u_t$$

- ▶ Demand System:

$$\Delta p_t = \alpha + \lambda_{v,t} \Delta \text{VIX}_t + \lambda_{F,t} \Delta \text{FF-Rate}_t + \dots + u_t$$

where $\lambda_{v,t}, \lambda_{F,t}$ are sums of investor-specific coefficients that are **time-varying**.⁴

- ▶ Does the demand system improve the fit?

⁴E.g. because of variations in the size distribution

Limitations of the linearized framework

- ▶ The paper is very rich and has a lot of cool results. What are the **limitations** of the simple separation between demand shift and slope?
- 1. Elasticity independent of *who* traded/caused price move
 - ▶ Investors can only observe the price, not who moved it.
 - ▶ Seems unrealistic at quarterly frequency. E.g. If hedge funds have demand shock (observed in futures market), perhaps investors respond differently.
- 2. Constant elasticities over time/regimes
 - ▶ What if elasticity different every quarter? Impossible to separate demand shock trade $u_{i,t}$ from elasticity trade $\zeta_{i,t}\Delta p_t$.
 - ▶ You actually show that ζ_i vary strongly across regimes...
- 3. Elasticity is independent of size of price move
 - ▶ Empirically, Fed steps in when things get out of control (March 2020).
 - ▶ Suggests concave price impact.⁵

⁵See Chaudhary & Li (2024) and many papers on square root law of market impact (e.g. Frazzini et al, 2018).

Conclusion

- ▶ Very nice paper!
- ▶ Important contribution to growing literature: Understanding asset markets (here: Treasuries) from the **bottom up** as opposed to running top-down regressions of prices onto macro-variables.

Comments:

- ▶ The paper effectively produces an accounting decomposition of yields by investor types. Make public and let people use it to test models/enrich empirics!
- ▶ For this it is useful to understand: What can potentially go wrong?